Exercise 10.1

Part B: Creative Questions

3.

a) If $\vec{a} = 2\vec{i} + 3\vec{j} + 5\vec{k}$, $\vec{b} = 7\vec{i} + 8\vec{j} - 6\vec{k}$, find $(\vec{a} + \vec{b}).(\vec{a} - \vec{b})$

Soln,

Given,

$\vec{a} = 2\vec{i} + 3\vec{j} + 5\vec{k}$

$\vec{b} = 7\vec{i} + 8\vec{j} - 6\vec{k}$

We have,

$\vec{a} + \vec{b} = (2\vec{i} + 3\vec{j} + 5\vec{k}) + (7\vec{i} + 8\vec{j} - 6\vec{k})$

$= 9\vec{i} + 11\vec{j} - \vec{k}$.

And,

$\vec{a} - \vec{b} = (2\vec{i} + 3\vec{j} + 5\vec{k}) - (7\vec{i} + 8\vec{j} - 6\vec{k})$

$= -5\vec{i} - 5\vec{j} + 11\vec{k}$

Now,

$(\vec{a} + \vec{b}).(\vec{a} - \vec{b}) = (9\vec{i} + 11\vec{j} - \vec{k}).(-5\vec{i} - 5\vec{j} + 11\vec{k})$

$= -45 - 55 + 11$

$= -111$

b) If $\vec{a} = (1,2,3)$ and $\vec{b} = (3,-1,2)$ then prove that $\vec{a} + \vec{b}$ and $\vec{a} - \vec{b}$ are orthogonal.

Soln,

$\vec{a} + \vec{b} = (1,2,3) + (3,-1,2)$

$= (4,1,5)$

$\vec{a} - \vec{b} = (1,2,3) - (3,-1,2)$

$= (-2,3,1)$

Now,

$(\vec{a} + \vec{b}).(\vec{a} - \vec{b}) = (4,1,5).(-2,3,1)$

$= -8 + 3 + 5$

$= -8 + 8$

$= 0$

Since, scalar product of $(\vec{a}+\vec{b})$ and $(\vec{a}-\vec{b})$ is zero so, $\vec{a}+\vec{b}$ and $\vec{a}-\vec{b}$ are orthogonal.

1. Find $\vec{a}\cdot\vec{b}$ if

a) $\vec{a} = 53\vec{i} + \vec{j} + 3\vec{k}$ and $\vec{b} = 3\vec{i} - 4\vec{j} + 7\vec{k}$

↳ Soln,

$\vec{a}\cdot\vec{b} = (53\vec{i} + \vec{j} + 3\vec{k}).(3\vec{i} - 4\vec{j} + 7\vec{k})$

$= 159 - 4 + 21$

$= 176$

b) $\vec{a} = 3\vec{i} + 2\vec{j} + \vec{k}$ and $\vec{b} = 4\vec{i} - \vec{j} + 3\vec{k}$

↳ Soln,

$\vec{a}\cdot\vec{b} = (3\vec{i} + 2\vec{j} + \vec{k}) \cdot (4\vec{i} - \vec{j} + 3\vec{k})$

$= 12 - 2 + 3$

$= 13$

2. Prove that the following pairs of vectors are orthogonal (perpendicular) to one another.

a) $\vec{a} = \vec{i} - 2\vec{j} + 5\vec{k}$ and $\vec{b} = -2\vec{i} + 4\vec{j} + 2\vec{k}$

↳ Soln,

$\vec{a}\cdot\vec{b} = (\vec{i} - 2\vec{j} + 5\vec{k}).(-2\vec{i} + 4\vec{j} + 2\vec{k})$

$= -2 - 8 + 10$

$= -10 + 10$

$= 0$

Since, $\vec{a}\cdot\vec{b} = 0$

$\therefore$ $\vec{a}$ and $\vec{b}$ are orthogonal with each other.

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b) $\vec{a} = (2,3,-5)$ and $\vec{b} = (2,2,2)$

↳ Soln,

$\vec{a}\cdot\vec{b} = (2,3,-5).(2,2,2)$

$= 4+6-10$

$= 10-10$

$= 0$

Since, $\vec{a}\cdot\vec{b} = 0$

$\therefore$ $\vec{a}$ and $\vec{b}$ are orthogonal with each other.

4. Prove that the vectors $\vec{a} = 6\vec{i} + 3\vec{j} + 2\vec{k}$, $\vec{b} = 2\vec{i} - 6\vec{j} + 3\vec{k}$ and $\vec{c} = -3\vec{i} + 2\vec{j} + 6\vec{k}$ are mutually perpendicular.

↳ Soln,

$\vec{a}\cdot\vec{b} = (6\vec{i}+3\vec{j}+2\vec{k}).(2\vec{i}-6\vec{j}+3\vec{k})$

$= 12 - 18 + 6$

$= 18-18$

$= 0$

$\vec{a}\cdot\vec{c} = (6\vec{i}+3\vec{j}+2\vec{k}).(-3\vec{i}+2\vec{j}+6\vec{k})$

$= -18+6+12$

$= -18+18$

$= 0$

$\vec{b}\cdot\vec{c} = (2\vec{i}-6\vec{j}+3\vec{k}).(-3\vec{i}+2\vec{j}+6\vec{k})$

$= -6-12+18$

$= -18+18$

$= 0$

5. a) Show that the vectors $2\vec{i}+3\vec{j}+\vec{k}$, $4\vec{i}+5\vec{j}+\vec{k}$ and $3\vec{i}+6\vec{j}-3\vec{k}$ are the vertices of a right-angled triangle.

↳ Soln,

let $\left.\begin{aligned} \vec{OA} &= 2\vec{i}+3\vec{j}+\vec{k} \\ \vec{OB} &= 4\vec{i}+5\vec{j}+\vec{k} \\ \vec{OC} &= 3\vec{i}+6\vec{j}-3\vec{k} \end{aligned}\right\}$ are vertices of a triangle ABC respectively.

Now,

$\cancel{|\vec{OC}|^2 = |\vec{OA}|^2 + |\vec{OB}|^2}$

$\cancel{\Rightarrow \left(\sqrt{3^2+6^2+(-3)^2}\right)^2 = \left(\sqrt{2^2+3^2+1^2}\right)^2 + \left(\sqrt{4^2+5^2+1^2}\right)^2}$

$\cancel{\Rightarrow \left(\sqrt{9+36+9}\right)^2 = \left(\sqrt{4+9+1}\right)^2 + \left(\sqrt{16+25+1}\right)^2}$

$\cancel{\Rightarrow 54 = 144}$

$\vec{AB} = \vec{OB} - \vec{OA}$

$= 4\vec{i}+5\vec{j}+\vec{k} - 2\vec{i} - 3\vec{j} - \vec{k}$

$= 2\vec{i}+2\vec{j}$

$\vec{BC} = \vec{OC} - \vec{OB}$

$= 3\vec{i}+6\vec{j}-3\vec{k} - 4\vec{i} - 5\vec{j} - \vec{k}$

$= -\vec{i}+\vec{j}-4\vec{k}$

$\vec{AB}\cdot\vec{BC} = (2\vec{i}+2\vec{j}).(-\vec{i}+\vec{j}-4\vec{k})$

$= -2+2+0$

$= 0$

Since $\vec{AB}\cdot\vec{BC} = 0$ i.e. $\vec{AB} \perp \vec{BC}$

$2\vec{i}+3\vec{j}+\vec{k}$, $4\vec{i}+5\vec{j}+\vec{k}$ and $3\vec{i}+6\vec{j}-3\vec{k}$ are the vertices of a right angled triangle

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6. Find the value of the scalar m if the following pairs of vectors are perpendicular.

a) $\vec{a} = m\vec{i} - 3\vec{j} + 5\vec{k}$, $\vec{b} = -m\vec{i} + m\vec{j} + 2\vec{k}$

Soln,

$\vec{a} \cdot \vec{b} = 0$ [For perpendicular).

$\Rightarrow (m\vec{i} - 3\vec{j} + 5\vec{k}) \cdot (-m\vec{i} + m\vec{j} + 2\vec{k}) = 0$

$\Rightarrow -m^2 - 3m + 10 = 0$

$\Rightarrow m^2 + 3m - 10 = 0$

$\Rightarrow m^2 + 5m - 2m - 10 = 0$

$\Rightarrow m(m+5) - 2(m+5) = 0$

$\Rightarrow (m-2)(m+5) = 0$

Either,

$m - 2 = 0$ $\quad$ $m + 5 = 0$

$m = 2$ $\quad$ $m = -5$

$\therefore m = -5, 2$

b) $\vec{a} = \vec{i} - 2\vec{j} + 4\vec{k}$, $\vec{b} = 2\vec{i} + 7\vec{j} + m\vec{k}$

Soln,

For perpendicular,

$\vec{a} \cdot \vec{b} = 0$

$\Rightarrow (\vec{i} - 2\vec{j} + 4\vec{k}) \cdot (2\vec{i} + 7\vec{j} + m\vec{k}) = 0$

$\Rightarrow 2 - 14 + 4m = 0$

$\Rightarrow 4m = 12$

$\Rightarrow m = \frac{\cancel{12}\ 3}{\cancel{4}}$

$\therefore m = 3$

7. Find the angle between the following pairs of vector.

a) $\vec{a} = \vec{i} + \vec{j} - 2\vec{k}$, $\vec{b} = 2\vec{i} - \vec{j} - \vec{k}$

Soln,

$|\vec{a}| = \sqrt{1^2 + 1^2 + (-2)^2}$

$= \sqrt{1+1+4}$

$= \sqrt{6}$

$|\vec{b}| = b = \sqrt{2^2 + (-1)^2 + (-1)^2}$

$= \sqrt{4+1+1}$

$= \sqrt{6}$

$\vec{a} \cdot \vec{b} = (\vec{i} + \vec{j} - 2\vec{k}) \cdot (2\vec{i} - \vec{j} - \vec{k})$

$= 2 - 1 + 2$

$= 3$

Now,

$\vec{a} \cdot \vec{b} = |\vec{a}| \, |\vec{b}| \cdot \cos\theta$

$\Rightarrow 3 = \sqrt{6} \cdot \sqrt{6} \cdot \cos\theta$

$\Rightarrow 3 = 6 \cdot \cos\theta$

$\Rightarrow \frac{3}{6} = \cos\theta$

$\Rightarrow \cos\theta = \frac{1}{2}$

$\Rightarrow \theta = \cos^{-1}\left(\frac{1}{2}\right)$

$\Rightarrow \theta = 60° = \frac{\pi}{3}$

b) $\vec{a} = 2\vec{i} - 3\vec{j} + \vec{k}$, $\vec{b} = 3\vec{i} - \vec{j} + 2\vec{k}$

Soln,

$|\vec{a}| = a = \sqrt{2^2 + (-3)^2 + 1^2}$

$= \sqrt{4+9+1}$

$= \sqrt{14}$

$|\vec{b}| = b = \sqrt{3^2 + (-1)^2 + 2^2}$

$= \sqrt{9+1+4}$

$= \sqrt{14}$

$\vec{a}.\vec{b} = (2\vec{i} - 3\vec{j} + \vec{k}).(3\vec{i} - \vec{j} + 2\vec{k})$

$= 6 + 3 + 2$

$= 11$

Now,

$\vec{a}.\vec{b} = ab\cos\theta$

$\Rightarrow 11 = \sqrt{14} \cdot \sqrt{14} \cdot \cos\theta$

$\Rightarrow 11 = 14 \cdot \cos\theta$

$\Rightarrow \cos\theta = \frac{11}{14}$

$\therefore \theta = \cos^{-1}\left(\frac{11}{14}\right)$ //.

5.b) If A, B, C and D are points (2,3,4), (5,4,-1), (3,6,2) and (1,2,0), respectively prove that AB is perpendicular to CD.

Soln: $OA = (2,3,4)$

$OB = (5,4,-1)$

$OC = (3,6,2)$

$OD = (1,2,0)$

$\vec{AB} = \vec{OB} - \vec{OA}$

$= (5,4,-1) - (2,3,4)$

$= (3,1,-5)$

$\vec{CD} = \cancel{(3,6,2)}\ \vec{OD} - \vec{OC}$

$= (1,2,0) - (3,6,2)$

$= (-2,-4,-2)$

Now,

$\vec{AB} \cdot \vec{CD} = (3,1,-5) \cdot (-2,-4,-2)$

$= -6 - 4 + 10$

$= -10 + 10$

$= 0$

This proves that AB is perpendicular to CD.

8. Given four points $A(2,-3,0)$, $B(1,-4,-2)$, $C(7,0,10)$ and $D(4,6,8)$. Then, find

a) the projection of AB on CD

→ Soln,

$\vec{AB} = \vec{OB} - \vec{OA}$
$= (1,-4,-2) - (2,-3,0)$
$= (-1,-1,-2)$

$\vec{CD} = \vec{OD} - \vec{OC}$
$= (4,6,8) - (7,0,10)$
$= (-3,6,-2)$

$|\vec{AB}| = \sqrt{(-1)^2 + (-1)^2 + (-2)^2}$
$= \sqrt{1+1+4}$
$= \sqrt{6}$

$|\vec{CD}| = \sqrt{(-3)^2 + 6^2 + (-2)^2}$
$= \sqrt{9+36+4}$
$= \sqrt{49}$
$= 7$

$\vec{AB} \cdot \vec{CD} = |\vec{CD}|$. projection of AB on CD

$\Rightarrow (-1,-1,-2) \cdot (-3,6,-2) = 7$. projection of AB on CD

$\Rightarrow 3 - 6 + 4 = 7$. projection of AB on CD

$\Rightarrow \frac{1}{7}$ = projection of AB on CD.

$\therefore$ projection of AB on CD $= \frac{1}{7}$ //.

b) projection of CD on $\overline{AB}$.

$\Rightarrow \vec{AB} \cdot \vec{CD} = 1$

Now,

$\vec{AB} \cdot \vec{CD} = |\vec{AB}|$ . projection of CD on AB

$\Rightarrow 1 = \sqrt{6}$ . projection of CD on AB

$\therefore$ projection of CD on AB $= \frac{1}{\sqrt{6}}$ //.

9.

b) prove that the angle between the vectors $\vec{a} = -2\vec{i} - \vec{k}$ and $\vec{b} = \vec{i} + \vec{j} - 3\vec{k}$ is given by $\cos\theta = \frac{1}{\sqrt{55}}$. Also, find the projection of one vector on direction of other.

Soln,

Given,

$\vec{a} = -2\vec{i} - \vec{k}$ and $\vec{b} = \vec{i} + \vec{j} - 3\vec{k}$

we have,

$\vec{a} \cdot \vec{b} = (-2\vec{i} - \vec{k}) \cdot (\vec{i} + \vec{j} - 3\vec{k})$

$= -2 + 0 + 3$

$= 1$

$|\vec{a}| = \sqrt{(-2)^2 + 0^2 + (-1)^2} = \sqrt{5}$

$|\vec{b}| = \sqrt{1^2 + 1^2 + (-3)^2} = \sqrt{11}$

Let $\theta$ be the angle between

$$\cos\theta = \frac{\vec{a}\cdot\vec{b}}{|\vec{a}|\,|\vec{b}|}$$

$$\text{or, } \cos\theta = \frac{1}{\sqrt{5}\cdot\sqrt{11}}$$

$$\text{or, } \cos\theta = \frac{1}{\sqrt{55}}$$ proved.

Again,

Projection of $\vec{a}$ on $\vec{b} = \frac{\vec{a}\cdot\vec{b}}{|\vec{b}|} = \frac{1}{\sqrt{11}}$

Projection of $\vec{b}$ on $\vec{a} = \frac{\vec{a}\cdot\vec{b}}{|\vec{a}|} = \frac{1}{\sqrt{5}}$

10.a) Show that the angles of the triangle with vertices $(1,-1,1)$, $(2,3,-1)$ and $(3,0,2)$ are $\cos^{-1}\frac{2}{\sqrt{114}}$, $\cos^{-1}\frac{4}{\sqrt{126}}$ and $\cos^{-1}\frac{17}{\sqrt{399}}$ respectively.

→ Soln,

Let the given triangle is ABC. So,

$\vec{OA} = (1,-1,1)$

$\vec{OB} = (2,3,-1)$

$\vec{OC} = (3,0,2)$

Where O is the origin.

We have,

$$\vec{AB} = \vec{OB} - \vec{OA}$$
$$= (2,3,-1) - (1,-1,1)$$
$$= (1,4,-2)$$

$$\vec{AC} = \vec{OC} - \vec{OA}$$
$$= (3,0,2) - (1,-1,1)$$
$$= (2,1,1)$$

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$\vec{BC} = \vec{OC} - \vec{OB}$
$= (3,0,2) - (2,3,-1)$
$= (1,-3,3)$

$|\vec{AB}| = \sqrt{1^2+4^2+(-2)^2} = \sqrt{21}$
$|\vec{AC}| = \sqrt{2^2+1^2+1^2} = \sqrt{6}$
$|\vec{BC}| = \sqrt{1^2+(-3)^2+3^2} = \sqrt{19}$

For angle A:
$\vec{AB} \cdot \vec{AC} = (1,4,-2) \cdot (2,1,1)$
$= 2+4-2$
$= 4$

$\therefore \cos A = \dfrac{\vec{AB} \cdot \vec{AC}}{|\vec{AB}| \cdot |\vec{AC}|}$

or, $\cos A = \dfrac{4}{\sqrt{21} \cdot \sqrt{6}}$

or, $\cos A = \dfrac{4}{\sqrt{126}}$

$\therefore A = \cos^{-1}\left(\dfrac{4}{\sqrt{126}}\right)$

For Angle B:
$\vec{BC} \cdot \vec{BA} = (1,-3,3) \cdot (-1,-4,2)$
$= -1+12+6$
$= 17$

$\therefore \cos B = \dfrac{\vec{BC} \cdot \vec{BA}}{|\vec{BC}| \cdot |\vec{BA}|}$

$\cos B = \dfrac{17}{\sqrt{19} \cdot \sqrt{21}}$

$\cos B = \dfrac{17}{\sqrt{399}}$

$\therefore B = \cos^{-1}\left(\dfrac{17}{\sqrt{399}}\right)$

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For angle C

$$\vec{CB} \cdot \vec{CA} = -\vec{BC} \cdot -\vec{AC}$$
$$= \vec{BC} \cdot \vec{AC}$$
$$= (1,-3,3) \cdot (2,1,1)$$
$$= 2 - 3 + 3$$
$$= 2$$

$$\therefore \cos C = \frac{-\vec{BC} \cdot -\vec{AC}}{|\vec{BC}| \cdot |\vec{AC}|}$$

$$\cos C = \frac{2}{\sqrt{19} \cdot \sqrt{6}}$$

$$\cos C = \frac{2}{\sqrt{144}}$$

$$\therefore C = \cos^{-1}\left(\frac{2}{\sqrt{144}}\right)$$

9.a) Find the cosine of the angle between the vectors $\vec{a} = 2\vec{i} - 3\vec{j} - 6\vec{k}$ and $\vec{b} = 2\vec{i} + 2\vec{j} - \vec{k}$. Also, find the projection of $\vec{a}$ on $\vec{b}$ and the projection of $\vec{b}$ on $\vec{a}$.

↪ Soln,

$$\vec{a} = 2\vec{i} - 3\vec{j} - 6\vec{k}$$
$$\vec{b} = 2\vec{i} + 2\vec{j} - \vec{k}$$

$$\vec{a} \cdot \vec{b} = (2\vec{i} - 3\vec{j} - 6\vec{k}) \cdot (2\vec{i} + 2\vec{j} - \vec{k})$$
$$= 4 - 6 + 6$$
$$= 4$$

$$|\vec{a}| = \sqrt{2^2 + (-3)^2 + (-6)^2}$$
$$= \sqrt{4 + 9 + 36}$$
$$= \sqrt{49} = 7$$

$$|\vec{b}| = \sqrt{2^2 + 2^2 + (-1)^2}$$
$$= \sqrt{4 + 4 + 1}$$

$= \sqrt{9} = 3$

Let $\theta$ be the angle between $\vec{a}$ and $\vec{b}$

So,

$$\cos\theta = \frac{\vec{a}\cdot\vec{b}}{|\vec{a}|.|\vec{b}|}$$

$$= \frac{4}{7\times 3}$$

$$= \frac{4}{21}$$

Again,

projection of $\vec{a}$ on $\vec{b} = \frac{\vec{a}\cdot\vec{b}}{|\vec{b}|}$

$$= \frac{4}{3}$$

Similarly,

projection of $\vec{b}$ on $\vec{a} = \frac{\vec{a}\cdot\vec{b}}{|\vec{a}|}$

$$= \frac{4}{7}$$

10.b) The position vectors of the vertices of a $\triangle ABC$ are $\vec{i} + 2\vec{j} + 4\vec{k}$ ; $-2\vec{i} + 2\vec{j} + \vec{k}$ and $2\vec{i} + 4\vec{j} - 3\vec{k}$. prove that the triangle is a right-angled triangle and find the other two angles of the triangle.

↳ Soln,

$\vec{OA} = \vec{i} + 2\vec{j} + 4\vec{k}$

$\vec{OB} = -2\vec{i} + 2\vec{j} + \vec{k}$

$\vec{OC} = 2\vec{i} + 4\vec{j} - 3\vec{k}$

We have,

$\vec{AB} = \vec{OB} - \vec{OA} = -2\vec{i} + 2\vec{j} + \vec{k} - \vec{i} - 2\vec{j} - 4\vec{k}$

$= -3\vec{i} - 3\vec{k}$

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$\vec{AC} = \vec{OC} - \vec{OA}$
$= 2\vec{i} + 4\vec{j} - 3\vec{k} - \vec{i} - 2\vec{j} - 4\vec{k}$
$= \vec{i} + 2\vec{j} - 7\vec{k}$

$\vec{BC} = \vec{OC} - \vec{OB}$
$= 2\vec{i} + 4\vec{j} - 3\vec{k} + 2\vec{i} - 2\vec{j} - \vec{k}$
$= 4\vec{i} + 2\vec{j} - 4\vec{k}$

Now,
$\vec{AB} \cdot \vec{BC} = (-3\vec{i} - 3\vec{k}) \cdot (4\vec{i} + 2\vec{j} - 4\vec{k})$
$= -12 + 12$
$= 0$

Since, $\vec{AB} \cdot \vec{BC} = 0$, The triangle is right angled triangle.

Again,
For Angle A,
$\vec{AB} \cdot \vec{AC} = (-3\vec{i} - 3\vec{k}) \cdot (\vec{i} + 2\vec{j} - 7\vec{k})$
$= -3 + 21$
$= 18$

~~Cos A =~~ $|\vec{AB}| = \sqrt{(-3)^2 + (-3)^2}$
$= \sqrt{9 + 9}$
$= \sqrt{18}$

$|\vec{AC}| = \sqrt{1^2 + 2^2 + (-7)^2}$
$= \sqrt{1 + 4 + 49}$
$= \sqrt{54}$

$|\vec{BC}| = \sqrt{4^2 + 2^2 + (-4)^2}$
$= \sqrt{16 + 4 + 16}$
$= \sqrt{36}$
$= 6$

$$\cos A = \frac{\vec{AB} \cdot \vec{AC}}{|\vec{AB}| \cdot |\vec{AC}|}$$

$$= \frac{18}{\sqrt{18} \times \sqrt{54}}$$

$$= \frac{1}{\sqrt{3}}$$

$$A = \cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

$$\therefore A \approx 54.73^\circ$$

For $\angle C$

$$\angle A + \angle B + \angle C = 180^\circ$$

$$\Rightarrow 54.73^\circ + 90^\circ + \angle C = 180^\circ$$

$$\Rightarrow \angle C \approx 35.27^\circ$$

11. If $\hat{a}$ and $\hat{b}$ are unit vectors and $\theta$ is the angle between them, prove that:

b) $\sin \theta/_2 = \frac{1}{2} |\hat{a} - \hat{b}|$

→ Soln,

To prove: $\sin \frac{\theta}{2} = \frac{1}{2} |\hat{a} - \hat{b}|$

Given, $\hat{a}$ and $\hat{b}$ are unit vectors and $\theta$ be the angle between them.

i.e. $|\hat{a}|$ and $|\hat{b}| = 1$.

Now,

$$(\hat{a} - \hat{b})^2 = (\hat{a})^2 - 2\hat{a} \cdot \hat{b} + (\hat{b})^2$$

$$|\hat{a} - \hat{b}|^2 = |\hat{a}|^2 - 2|\hat{a}||\hat{b}| \cos\theta + |\hat{b}|^2$$

$$(\because |\vec{a}|^2 = \vec{a}^2)$$

or, $|\hat{a} - \hat{b}|^2 = 1 - 2\cos\theta + 1$

or, $|\hat{a} - \hat{b}|^2 = 2 - 2\cos\theta$

or, $|\hat{a} - \hat{b}|^2 = 2(1 - \cos\theta)$

or, $|\hat{a} - \hat{b}|^2 = 2 \times 2 \sin^2 \frac{\theta}{2}$

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or, $|\hat{a}-\hat{b}|^2 = \left(2\sin\frac{\theta}{2}\right)^2$

or, $|\hat{a}-\hat{b}| = 2\sin\frac{\theta}{2}$

$\Rightarrow \sin\frac{\theta}{2} = \frac{1}{2}|\hat{a}-\hat{b}|$

Proved

a) $\cos\frac{\theta}{2} = \frac{1}{2}|\hat{a}+\hat{b}|$

→ To prove: $\cos\theta/2 = \frac{1}{2}|\hat{a}+\hat{b}|$

Soln,

$(\hat{a}+\hat{b})^2 = (\hat{a})^2 + 2\hat{a}\cdot\hat{b} + (\hat{b})^2$

$|\hat{a}+\hat{b}|^2 = |\hat{a}|^2 + 2\hat{a}\cdot\hat{b} + |\hat{b}|^2$

$\Rightarrow |\hat{a}+\hat{b}|^2 = |\hat{a}|^2 + 2|\hat{a}||\hat{b}|\cos\theta + |\hat{b}|^2$

$\Rightarrow |\hat{a}+\hat{b}|^2 = 1 + 2\cos\theta + 1$

$\Rightarrow |\hat{a}+\hat{b}|^2 = 2(1+\cos\theta)$

$\Rightarrow |\hat{a}+\hat{b}|^2 = 2 \times 2\cos^2\frac{\theta}{2}$

$\Rightarrow |\hat{a}+\hat{b}|^2 = \left(2\cos\frac{\theta}{2}\right)^2$

$|\hat{a}+\hat{b}| = 2\cos\frac{\theta}{2}$

$\cos\frac{\theta}{2} = \frac{1}{2}|\hat{a}+\hat{b}|$

$\therefore \cos\frac{\theta}{2} = \frac{1}{2}|\hat{a}+\hat{b}|$

Proved

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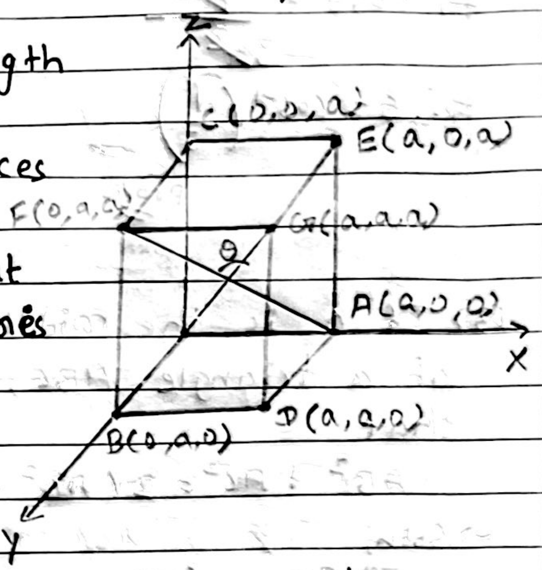
12. Prove that the angle between any two diagonals of a cube is $\cos^{-1}\left(\frac{1}{3}\right)$.

→ Soln,

Consider a cube of length 'a' as shown in figure. Suppose one of the vertices is at origin and three adjacent edges from that vertex be x, y and z-axes respectively.




The co-ordinates of vertices are shown in the figure.

Let us consider two diagonals OG and AF.

Now,

$\vec{OG} = (a, a, a)$ and,

$\vec{AF} = \vec{OF} - \vec{OA}$

$= (0,a,a) - (a,0,0)$

$= (-a, a, a)$

$\vec{OG} \cdot \vec{AF} = (a, a, a) \cdot (-a, a, a)$

$= -a^2 + a^2 + a^2$

$= a^2$

$|\vec{OG}| = \sqrt{a^2+a^2+a^2} = \sqrt{3a^2} = a\sqrt{3}$

$|\vec{AF}| = \sqrt{(-a)^2+a^2+a^2} = a\sqrt{3}$

Let θ be the angle between the diagonals.

So,

$\cos\theta = \dfrac{\vec{OG} \cdot \vec{AF}}{|\vec{OG}| \cdot |\vec{AF}|}$

$$\Rightarrow \cos\theta = \frac{a^2}{a\sqrt{3} \times a\sqrt{3}}$$

$$\Rightarrow \cos\theta = \frac{\cancel{a^2}}{3\cancel{a^2}}$$

$$\Rightarrow \therefore \theta = \cos^{-1}\left(\frac{1}{3}\right)$$

proved

13) If O is the middle point of the side BC of a triangle ABC, prove, by vector method, that

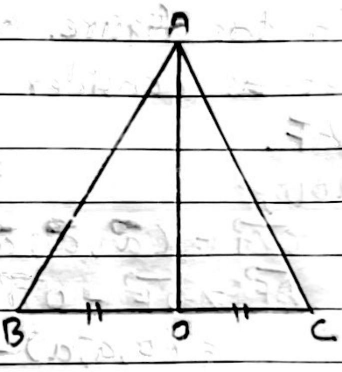
$$AB^2 + AC^2 = 2(AO^2 + BO^2).$$

↳ Soln,

~~AB~~ From figure,

$\vec{AB} = \vec{AO} + \vec{OB}$

$\vec{AC} = \vec{AO} + \vec{OC}$




Now,

$$AB^2 + AC^2 = (\vec{OB} - \vec{OA})^2 + (\vec{OC} - \vec{OA})^2$$

$\because OC = BO$

$$AB^2 + AC^2 = (\vec{OB} - \vec{OA})^2 + (\vec{BO} - \vec{OA})^2$$
$$= (\vec{OB} - \vec{OA})^2 + (-\vec{OB} - \vec{OA})^2$$
$$= (\vec{OB} - \vec{OA})^2 + (\vec{OB} + \vec{OA})^2$$
$$= |\vec{OB}|^2 - \cancel{2\vec{OB}\cdot\vec{OA}} + |\vec{OA}|^2 + |\vec{OB}|^2 + \cancel{2\vec{OB}\cdot\vec{OA}} + |\vec{OA}|^2$$
$$= 2(|\vec{OB}|^2 + |\vec{OA}|^2)$$
$$= 2(|\vec{BO}|^2 + |\vec{AO}|^2)$$
$$= 2(BO^2 + AO^2) \quad (\because |\vec{BO}|^2 = BO^2)$$

14. By using vector method, prove that:

a) if the diagonals of a parallelogram are equal,

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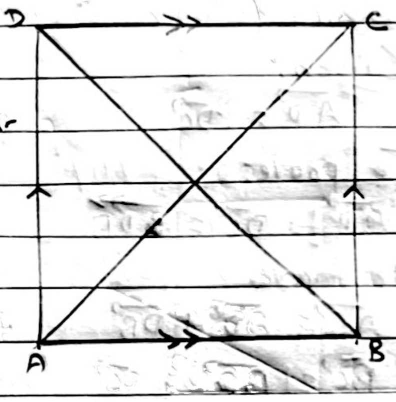
then it is a rectangle.




→ Soln,

Given, here ABCD is a parallelogram in which the diagonals AC and BD are equal.

To prove: ABCD is a rectangle.

We have,

$AC = BD$

or, $AC^2 = BD^2$ ( ∵ Squaring on both sides)

or, $\vec{AC}^2 = \vec{BD}^2$ ( ∵ $\vec{a}^2 = |\vec{a}|^2$ )

or, $(\vec{AB} + \vec{BC})^2 = (\vec{BC} + \vec{CD})^2$ ( ∵ by Triangle law of vector addition).

or, $(\vec{AB} + \vec{BC})^2 = (\vec{BC} + \vec{BA})^2$ ( ∵ $\vec{CD} = \vec{BA}$ )

or, $(\vec{BC} + \vec{AB})^2 = (\vec{BC} - \vec{AB})^2$

or, $\cancel{\vec{BC}^2} + 2\vec{AB}\cdot\vec{BC} + \cancel{\vec{AB}^2} = \cancel{\vec{BC}^2} - 2\vec{AB}\cdot\vec{BC} + \cancel{\vec{AB}^2}$

or, $4\vec{AB}\cdot\vec{BC} = 0$

or, $\vec{AB}\cdot\vec{BC} = 0$

i.e. $\vec{AB}$ is Perpendicular to $\vec{BC}$.

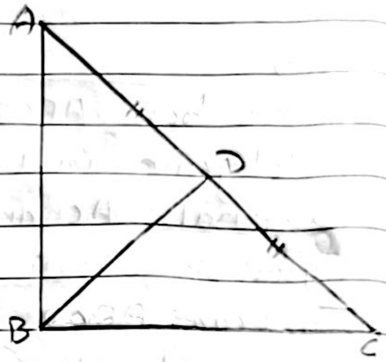
We know that, a parallelogram having 90° angle is a rectangle.

∴ ABCD is a rectangle.

Proved

b) The middle point of the hypotenuse of a right-angled triangle is equidistant from the vertices.

→ Soln,



Given,

$\vec{AD} = \vec{DC}$

To prove:

$\vec{AD} = \vec{DC} = \vec{DB}$

We have,

$\vec{AB} = \vec{AD} + \vec{DB}$

$\vec{BC} = \vec{BD} + \vec{DC}$

$= \vec{AD} - \vec{DB}$

Now,

$\vec{AB} \cdot \vec{BC} = 0$

or, $(\vec{AD} + \vec{DB}) \cdot (\vec{AD} - \vec{DB}) = 0$

or, $\vec{AD}^2 - \vec{DB}^2 = 0$

or, $AD^2 - DB^2 = 0$

or, $AD^2 = DB^2$

or, $AD = DB$

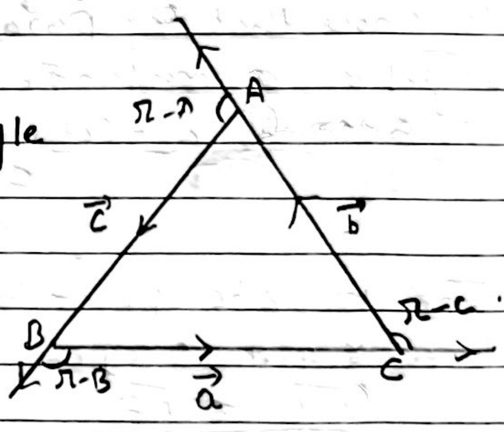
$\therefore AD = DC = DB$.

Q. In Any triangle ABC prove by using vector method that:

ⓐ $b^2 = a^2 + c^2 - 2ac\cos B$ [cosine Law]

ⓑ $a = b\cos C + c\cos B$ [projection Law]

→ Soln,



a) Let ABC be a triangle such that $\vec{AB} = \vec{c}$, $\vec{BC} = \vec{a}$, and $\vec{CA} = \vec{b}$. by triangle law of vector addition.

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$\vec{a}+\vec{b}+\vec{c}=0$

$\Rightarrow \vec{b} = -\vec{a}-\vec{c}$

$\Rightarrow \vec{b} = -(\vec{a}+\vec{c})$

squaring on both sides,

$\vec{b}^2 = (\vec{a}+\vec{c})^2$

$\vec{b}^2 = \vec{a}^2 + 2\vec{a}.\vec{c} + \vec{c}^2$

$\vec{b}^2 = \vec{a}^2 + 2\vec{a}\vec{c} + \vec{c}^2$

$b^2 = a^2 + 2\vec{a}.\vec{c} + c^2 \quad [\because \vec{a}^2 = a^2]$

The angle between $\vec{a}$ and $\vec{c}$ is $\pi - B$.

so,

$b^2 = a^2 + 2ac\cos(\pi - B) + c^2$

$\therefore b^2 = a^2 + c^2 - 2ac\cos B$

b)

Proof

Let ABC be a triangle as shown in figure where $\vec{AB}=\vec{c}$, $\vec{BC}=\vec{a}$ and $\vec{CA}=\vec{b}$.

By Triangle law of vector Addition,

$\vec{a}+\vec{b}+\vec{c}=0$

or, $\vec{a} = -\vec{b}-\vec{c}$

Taking dot product on both sides by $\vec{a}$.

We get,

$\vec{a}.\vec{a} = -\vec{a}.\vec{b} - \vec{a}.\vec{c}$

$\Rightarrow \vec{a}^2 = -\vec{a}.\vec{b} - \vec{a}.\vec{c}$

$\Rightarrow |\vec{a}|^2 = -\vec{a}.\vec{b} - \vec{a}.\vec{c}$

From figure, the angle between $\vec{a}$ & $\vec{b}$ is $\pi - C$ and between $\vec{a}.\vec{c} = \pi - B$. so,

$a^2 = -ab\cos(\pi - C) - ac\cos(\pi - B)$

$\Rightarrow a^2 = ab\cos C + ac\cos B$

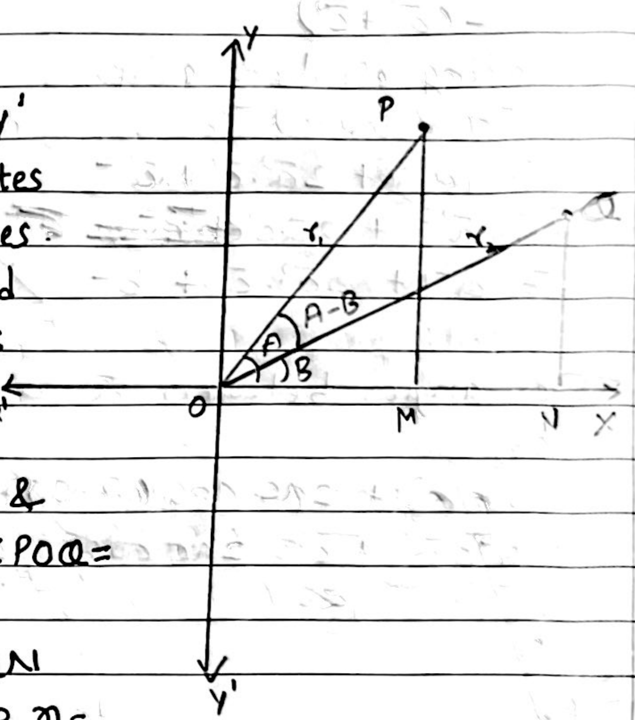
$\Rightarrow a = b\cos C + c\cos B$

Proved

# Prove by vector method :

$\cos(A-B) = \cos A \cdot \cos B + \sin A \cdot \sin B$

proof




Let $XOX'$ and $YOY'$ respectively denotes the x and Y-axes. ~~Let~~ Suppose P and Q be two points such that $\angle POX = A$. Similarly $\angle QOX = B$, $OP = r_1$ & $OQ = r_2$. then, $\angle POQ = A - B$.

Draw PM and QN perpendiculars to x-axis.

In right angled triangle POM,

$\sin A = \dfrac{PM}{OP} \Rightarrow \sin A = \dfrac{PM}{r_1}$

$\Rightarrow PM = r_1 \sin A$

And,

$\cos A = \dfrac{OM}{OP} \Rightarrow \cos A = \dfrac{OM}{r_1}$

$\Rightarrow OM = r_1 \cos A$

So, the co-ordinates of point $P = (OM, PM)$

$P = (r_1 \cos A, r_1 \sin A)$

$\therefore \vec{OP} = (r_1 \cos A, r_1 \sin A, 0)$

Similarly,

the co-ordinates of point $Q = (r_2 \cos B, r_2 \sin B)$

$\therefore \vec{OQ} = (r_2 \cos B, r_2 \sin B, 0)$

We have,

$$\overrightarrow{OP}\cdot\overrightarrow{OQ} = (r_1\cos A, r_1\sin A, 0)\cdot(r_2\cos B, r_2\sin B, 0)$$
$$= r_1 r_2 \cos A\cdot\cos B + r_1 r_2 \sin A\cdot\sin B + 0$$
$$= r_1 r_2 (\cos A\cdot\cos B + \sin A\cdot\sin B)$$

Now the angle between $\overrightarrow{OP}$ and $\overrightarrow{OQ}$ is $A-B$.

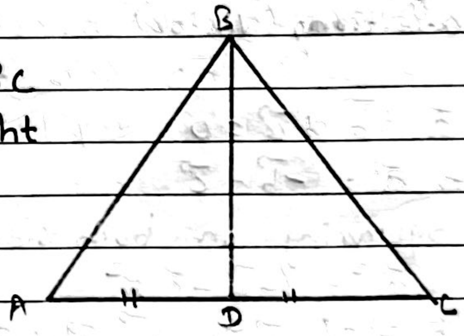
so,

$$\cos(A-B) = \frac{\overrightarrow{OP}\cdot\overrightarrow{OQ}}{|\overrightarrow{OP}|.|\overrightarrow{OQ}|}$$
$$= \frac{\cancel{r_1}\cancel{r_2}(\cos A\cdot\cos B + \sin A\cdot\sin B)}{\cancel{r_1}.\cancel{r_2}}$$

$\therefore \cos(A-B) = \cos A\cdot\cos B + \sin A\cdot\sin B$

Proved.

14.c) If a median of a triangle is perpendicular to the base, it is an isosceles triangle.




↳ let us consider a $\Delta ABC$ with base AC and height BD.

Given: $\overrightarrow{AD} = \overrightarrow{DC}$, $\overrightarrow{AC} \perp DB$

To prove: $AB = CB$

Proof

$$\overrightarrow{BD} = \frac{1}{2}(\overrightarrow{BA} + \overrightarrow{BC})$$

and $\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{BC}$

According to question,

$$\overrightarrow{AC}\cdot\overrightarrow{BD} = 0$$

$$\Rightarrow \frac{1}{2}(\overrightarrow{BC} - \overrightarrow{AB})\cdot(\overrightarrow{BC} + \overrightarrow{AB}) = 0$$

$$\Rightarrow \frac{1}{2}(\overrightarrow{BC}^2 - \overrightarrow{AB}^2) = 0$$

$$\Rightarrow \overrightarrow{BC}^2 - \overrightarrow{AB}^2 = 0$$

$\overrightarrow{BC}^2 = \overrightarrow{AB}^2$

$\therefore \overrightarrow{BC} = \overrightarrow{AB}$

$\therefore$ If a median of a triangle is perpendicular to the base, it is an isosceles triangle.

proved

15. In any $\Delta ABC$, prove vectorically that:

a) $a^2 = b^2 + c^2 - 2bc\cos A$

Let ABC be a triangle such that $\overrightarrow{AB} = \vec{c}$, $\overrightarrow{BC} = \vec{a}$, $\overrightarrow{AC} = \vec{b}$.

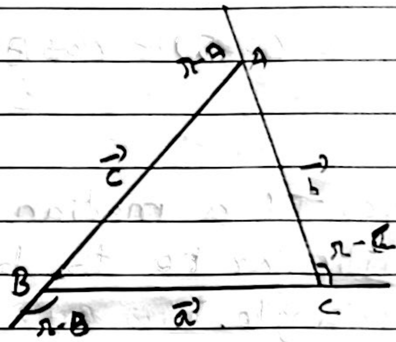


fig (i)

Now,

By triangle law of addition, (vector)

$\vec{a} + \vec{b} + \vec{c} = 0$

$\Rightarrow \vec{a} = -\vec{b} - \vec{c}$

Squaring on both sides,

$\vec{a}^2 = (-\vec{b} - \vec{c})^2$

$\Rightarrow \vec{a}^2 = \vec{b}^2 + 2\vec{b}\cdot\vec{c} + \vec{c}^2$

$\Rightarrow \vec{a}^2 = b^2 + 2bc\cos(\pi - A) + c^2$ [$\because \vec{a}^2 = a^2$]

$\Rightarrow a^2 = b^2 + c^2 - 2bc\cos A$

b) $c^2 = a^2 + b^2 - 2ab\cos C$

proof

By triangle law of vector addition.

$\vec{a} + \vec{b} + \vec{c} = 0$

$\Rightarrow \vec{c} = -\vec{a} - \vec{b}$

$\Rightarrow$ Squaring on both sides,

$\Rightarrow \vec{c}^2 = \{(\vec{a} + \vec{b})\}^2$

$\Rightarrow \vec{c}^2 = \vec{a}^2 + 2\vec{a}\cdot\vec{b} + \vec{b}^2$

$\Rightarrow c^2 = a^2 + 2\vec{a}\cdot\vec{b} + b^2 \quad [\because \vec{a}^2 = a^2]$

The angle between $\vec{a}$ and $\vec{b}$ is $(\pi - C)$ so,

$c^2 = a^2 + 2ab\cos(\pi - C) + b^2$

$\Rightarrow c^2 = a^2 - 2ab\cos C + b^2$

$\Rightarrow c^2 = a^2 + b^2 - 2ab\cos C$

proved

c) $b = c\cos A + a\cos C$

Proof

By vector law of Addition,

$\vec{a} + \vec{b} + \vec{c} = 0$

$\Rightarrow \vec{b} = -\vec{a} - \vec{c}$

Taking dot product on both sides by $\vec{b}$:

$\vec{b}\cdot\vec{b} = -\vec{a}\cdot\vec{b} - \vec{c}\cdot\vec{b}$

$\Rightarrow \vec{b}^2 = -\vec{a}\cdot\vec{b} - \vec{c}\cdot\vec{b}$

$\Rightarrow$ From figure (i) the angle between $\vec{a}$ & $\vec{b}$ is $\pi - C$ and $\vec{c}$ & $\vec{b}$ is $\pi - A$. so,

$b^2 = -ab\cos(\pi - C) - bc\cos(\pi - A) \quad [\because \vec{b}^2 = b^2]$

$\Rightarrow b^2 = ab\cos C + bc\cos A$

$\Rightarrow b^{\cancel{2}} = \cancel{b}(a\cos C + c\cos A)$

$\Rightarrow b = a\cos C + c\cos A$

proved

d) $c = a\cos B + b\cos A$

Proof

By vector law of addition,

$\vec{a} + \vec{b} + \vec{c} = 0$

$\vec{c} = -\vec{a} - \vec{b}$

Taking dot product on both sides by $\vec{c}$

$\vec{c}\cdot\vec{c} = -\vec{a}\cdot\vec{c} - \vec{b}\cdot\vec{c}$

$\vec{c}^{\,2} = -\vec{a}\cdot\vec{c} - \vec{b}\cdot\vec{c}$

$c^2 = -\vec{a}\cdot\vec{c} - \vec{b}\cdot\vec{c}$ $\quad [\because \vec{c}^{\,2} = c^2]$

From figure, The angle between $\vec{a}$ & $\vec{c}$ is $\pi - B$ and between $\vec{b}$ & $\vec{c}$ is $\pi - A$.

$\Rightarrow c^2 = -ac\cos(\pi - B) - bc\cos(\pi - A)$

$\Rightarrow c^2 = +ac\cos B + bc\cos A$

$\Rightarrow c^{\cancel{2}} = \cancel{c}(a\cos B + b\cos A)$

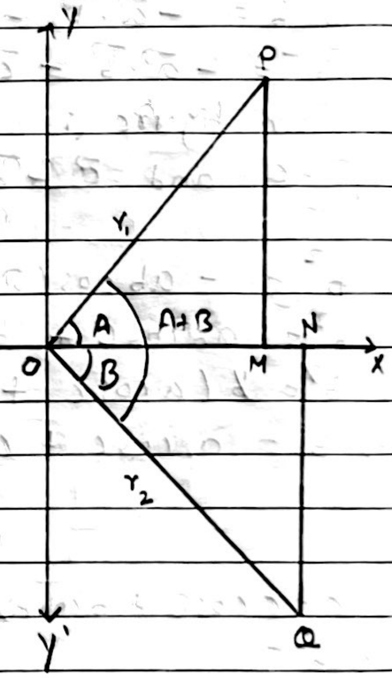
$\therefore c = a\cos B + b\cos A$ [proved]

16. Prove by vector method:

a) $\cos(A+B) = \cos A \cdot \cos B - \sin A \cdot \sin B$

↳ Soln,

Let XOX' and YOY' respectively denotes the X and Y-axes. Suppose P and Q be two points such that $\angle POX = A$. Similarly $\angle QOX = B$, $OP = r_1$ & $OQ = r_2$. then, $\angle POQ = A+B$.




Draw PM and QN perpendicular to x-axis.

In right angled $\triangle POM$,

$\sin A = \frac{PM}{OP} \Rightarrow \sin A = \frac{PM}{r_1}$

$\Rightarrow PM = r_1 \sin A$

And,

$\cos A = \frac{OM}{OP} \Rightarrow \cos A = \frac{OM}{r_1}$

$\Rightarrow OM = r_1 \cos A$

So, the co-ordinates of point $P = (OM, PM)$

$P = (r_1 \cos A, r_1 \sin A)$

$\therefore \vec{OP} = (r_1 \cos A, r_1 \sin A, 0)$

Similarly,

the co-ordinates of point $Q = (r_2 \cos B, -r_2 \sin B)$

negative sign indicates that co-ordinates lies at opposite axis.

i.e. $4^{th}$ quadrant

We have,

$\vec{OP} \cdot \vec{OQ} = (r_1 \cos A, r_1 \sin A, 0) \cdot (r_2 \cos B, -r_2 \sin B, 0)$

$= r_1 r_2 \cos A \cdot \cos B - r_1 r_2 \sin A \cdot \sin B + 0$

$= r_1 r_2 (\cos A \cdot \cos B - \sin A \cdot \sin B)$

Now the angle between $\vec{OP}$ and $\vec{OQ}$ is $A+B$.

So,

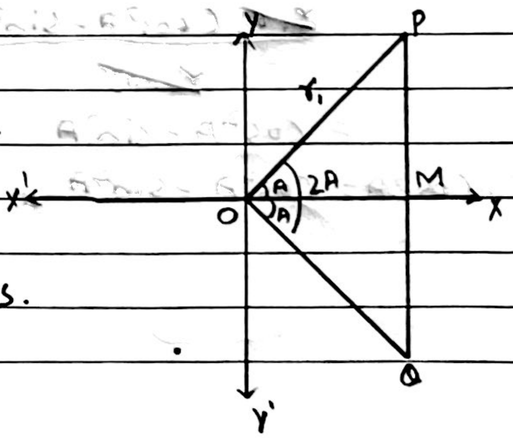
$\cos(A+B) = \frac{\vec{OP} \cdot \vec{OQ}}{|\vec{OP}| \cdot |\vec{OQ}|}$

$= \frac{\cancel{r_1}\cancel{r_2}(\cos A \cdot \cos B - \sin A \cdot \sin B)}{\cancel{r_1}\cancel{r_2}}$

$\therefore \cos(A+B) = \cos A \cdot \cos B - \sin A \cdot \sin B$

Proved.

b) $\cos 2A = \cos^2 A - \sin^2 A$




Let $XOX'$ and $YOY'$ respectively denotes the $X$ and $Y$-axes. Suppose $P$ and $Q$ be two points.

Such that $\angle POX = \angle QOX = A$, $OP = r_1$, $OQ = r_2$
then $\angle POQ = 2A$

Draw PM and QM perpendicular to $x$-axis.
In right angled $\Delta POM$

$\sin A = \frac{PM}{OP} \Rightarrow \sin A = \frac{PM}{r_1}$

$\Rightarrow PM = r_1 \sin A$

And,

$\cos A = \frac{OM}{OP} \Rightarrow \cos A = \frac{OM}{r_1}$

$\Rightarrow OM = r_1 \cos A$

So, the co-ordinates of point $P = (OM, PM)$
$\therefore P = (r_1 \cos A, r_1 \sin A)$

Similarly,
The co-ordinates of point $Q = (r_2 \cos A, -r_2 \sin A)$
Negative sign because point lies on 4th quad-rant.

We have,

$\vec{OP} \cdot \vec{OQ} = (r_1 \cos A, r_1 \sin A, 0) \cdot (r_2 \cos A, -r_2 \sin A, 0)$

$= r_1 r_2 \cos^2 A - r_1 r_2 \sin^2 A + 0$

$= r_1 r_2 (\cos^2 A - \sin^2 A)$

Now,
the angle between OP and OQ is 2A.
So,

$\cos 2A = \frac{\vec{OP} \cdot \vec{OQ}}{|\vec{OP}| \cdot |\vec{OQ}|}$

$= \frac{\cancel{r_1 r_2} (\cos^2 A - \sin^2 A)}{\cancel{r_1 \cdot r_2}}$

$= \cos^2 A - \sin^2 A$

$\therefore \cos 2A = \cos^2 A - \sin^2 A$